

# How Do Users Review Spotify? Analyzing Emotion, Topics, and Platform Replies Using Text-as-Data

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Significance Statement

Understanding how Spotify selectively replies to user reviews sheds light on platform governance, digital fairness, and customer engagement strategy. By analyzing emotional tone and review content using TAD and logistic regression, this study highlights patterns in review topics and corporate responsiveness, contributing to business insight and broader discussions on digital transparency and priority.

Abstract

This study analyzes 60,000 Spotify reviews to examine what drives platform replies. Using text classification, topic modeling, and logistic regression, we find replies are more likely for detailed, negative feedback—especially with technical issues or emotions like disgust. Joy, praise, and competitor mentions have little impact.

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1 Introduction

1.1 Motivation

Spotify has become the global leader in music streaming, with over 252 million premium subscribers as of Q3 2024 (Statista, 2024; Figure 1).

While subscription metrics reveal growth, understanding user satisfaction and retention requires deeper analysis beyond numerical figures. User-generated content (UGC), especially app store reviews, provides insights into customer experiences.

These reviews reflect not only product satisfaction but also user frustrations and unmet expectations. Platforms like Spotify occasionally respond to user reviews, but the dynamics of which reviews receive replies remain largely unexplored.

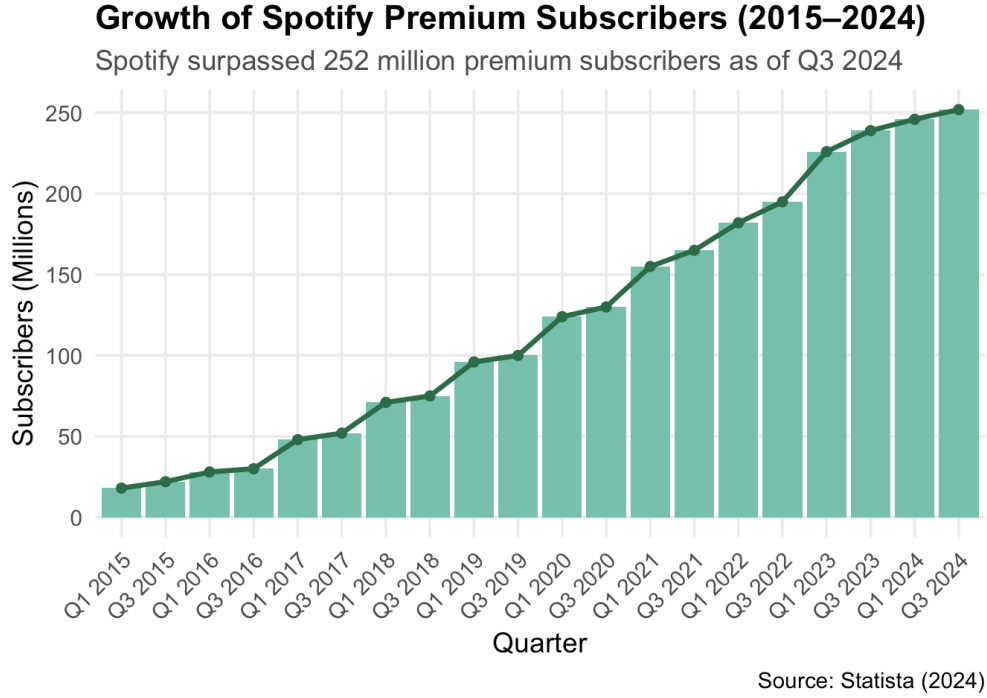


Figure 1. Number of Spotify premium subscribers worldwide from Q1 2015 to Q4 2024.

## 1.2 Prior Research

Literature supports that text analysis informs consumer insights. Pang & Lee (2008) and Liu (2012) laid the foundation of sentiment analysis for understanding customer satisfaction. Tirunillai & Tellis (2012) showed that user-generated content predicts business performance, while Blei et al. (2003) introduced LDA topic modeling to identify consumer concerns.

In digital media, Datta et al. (2018) found that users’ sentiments correlate with music subscriber growth, and Piskorski et al. (2019) demonstrated that negative sentiment spikes signal churn risk. These findings support the analysis of user sentiment in Spotify’s reviews and can inform market decisions.

## 1.3 Research Questions

- **RQ1:** How do Spotify users describe their experience in written reviews?
- **RQ2:** Which review topics or tones are more likely to receive an official reply—and what does this reveal about platform priorities or transparency?

## 1.4 Report Overview

This report begins by describing the dataset and annotation process, followed by supervised classification and topic modeling. I then evaluate emotional tone and reply likelihood through logistic regression, and conclude with a discussion of key findings and implications.

## 2 Data and Methods

### 2.1 Data Source and Unit of Observation

The dataset, published on Kaggle (2022), contains 61,594 Spotify app reviews from the Google Play Store. Each entry includes the text of the review, timestamp, rating (1–5), helpfulness score, and reply (if any). Each observation corresponds to one user’s review.

### 2.2 Variables of Interest

The dataset used in this study comprises a set of variables that support both classification and regression analysis (Table 1). Each review is accompanied by structured metadata and derived features that inform our analysis of user experience and reply behavior.

**Table 1. Variables of Interest in the Spotify Review Dataset**

| Variable Name      | Description                                                      |
|--------------------|------------------------------------------------------------------|
| Review             | Full text of the user review                                     |
| Rating             | Star rating (1 to 5 scale)                                       |
| Predicted_Label    | Binary label (App Experience vs. Other) via TF-IDF Lasso         |
| Sentiment          | Derived from rating: Negative (1–2), Neutral (3), Positive (4–5) |
| Replied            | Binary indicator if Spotify replied                              |
| PredictedLabel.LDA | Topic label from LDA topic modeling                              |
| NRC Emotion        | Ten emotion scores from NRC Lexicon                              |
| word.count         | Total number of words in review                                  |
| competitor.mention | Binary indicator if a competitor is mentioned                    |

### 2.3 Data Wrangling Steps

To prepare the dataset for analysis, I implemented a structured data cleaning and transformation pipeline.

First, I converted all review text to lowercase, removed duplicates, and eliminated special characters to ensure consistency. The cleaned text was then tokenized and matched with the NRC lexicon to compute review-level emotion scores across ten emotion categories.

Using a manually labeled random sample of 2,000 reviews, I initially created four functional categories: Technical Issues, UX & Features, Price & Ads, and Other. However, due to overlapping themes and classification ambiguity—especially between Technical and UX feedback—I consolidated these into a binary scheme to improve annotation consistency and model performance. Therefore, the final categories were: (1) “App Experience”: reviews mainly focused on technical problems, user interface, and app functionality; (2) “Other”: reviews related to pricing, ads, music content, or general sentiment. The “App Experience” category may include all forms of interaction with the Spotify platform, including the mobile app, website, and other interfaces. It may also cover reviews that partially mention pricing, praise, or subscription but are primarily focused on user

experience. This simplified structure allowed the classifier to more reliably distinguish actionable user feedback tied to user experience.

I also engineered several features to support downstream analysis. Binary flags were created to indicate whether a review received an official Spotify reply and whether a competing platform (e.g., Apple Music) was mentioned. I computed word counts for each review and detected specific competitor references.

For classification, I constructed both TF-IDF and word embedding matrices to predict topic labels. To perform unsupervised topic modeling, I filtered the App Experience reviews and preprocessed them for input into an LDA model.

To address class imbalance in reply behavior, I applied SMOTE (Synthetic Minority Oversampling Technique), generating synthetic samples for the minority class (Replied = 1). This rebalancing allowed me to fit more robust logistic regression models.

## 3 Analysis

### 3.1 Supervised Classification

To distinguish between App Experience and Other reviews, I trained three supervised models—Naïve Bayes, SVM, and Lasso—on the annotated subset. Among these, the Lasso logistic regression model achieved the highest accuracy (83%) and interpretability (Table 2). I applied this model to classify the remaining reviews in the dataset.

**Table 2. Performance Metrics of Lasso Models**

| Metric            | TF-IDF vs. Word Embedding |
|-------------------|---------------------------|
| Accuracy          | 0.83, 0.82                |
| Kappa             | 0.648, 0.613              |
| Precision         | 0.868, 0.837              |
| Recall            | 0.846, 0.878              |
| F1 Score          | 0.857, 0.857              |
| Balanced Accuracy | 0.826, 0.802              |

### 3.2 Topic Modeling

LDA was applied to App Experience reviews (~38,000 records), revealing six subtopics (Table 3): Technical Fixes, Account & Subscription, General UX Feedback, Playback Issues, Music & Playlist Features, and Positive Experience.

**Table 3. LDA-Inferred Topics for App Experience Reviews**

| Topic                       | Interpretation                                         |
|-----------------------------|--------------------------------------------------------|
| Technical Fixes             | Bugs, app malfunctions, connection issues              |
| Account & Subscription      | Login, billing, and premium access issues              |
| General UX Feedback         | Vague usability comments and minor frustrations        |
| Playback Issues             | Playback interruptions, especially after updates       |
| Music and Playlist Features | Playlist quality, song selection, and feature feedback |
| Positive Experience         | Expressions of satisfaction and brand loyalty          |

### 3.3 Sentiment and Emotion

Sentiment was categorized based on ratings: 1–2 as negative, 3 neutral, and 4–5 positive. NRC Lexicon provided 10 emotion scores (e.g., anger, joy, disgust) for each review.

### 3.4 Logistic Regression: Reply Prediction

I modeled reply likelihood using emotional scores, review length, competitor mentions, and topic labels. A second model applied SMOTE to account for class imbalance. The model specification is shown as follows:

$$\begin{aligned}
\text{logit}(\Pr(\text{Replied}_i = 1)) = & \beta_0 + \beta_1 \cdot \text{anticipation}_i + \beta_2 \cdot \text{anger}_i + \beta_3 \cdot \text{disgust}_i + \beta_4 \cdot \text{fear}_i \\
& + \beta_5 \cdot \text{joy}_i + \beta_6 \cdot \text{sadness}_i + \beta_7 \cdot \text{surprise}_i + \beta_8 \cdot \text{trust}_i \\
& + \beta_9 \cdot \text{negative}_i + \beta_{10} \cdot \text{positive}_i + \beta_{11} \cdot \text{word\_count}_i \\
& + \beta_{12} \cdot \text{competitor\_mention}_i + \beta_{13} \cdot \text{PredictedLabel\_LDA}_i + \varepsilon_i
\end{aligned}$$

## 4 Results

### 4.1 Descriptive Overview

49 percent of Spotify reviews are positive in tone, with around 60 percent classified under the App Experience category. However, only 3.5 percent of reviews received an official reply from the platform, highlighting a significant gap in engagement (Figure 2).

### 4.2 Sentiment, Emotion Score, and Reply

Spotify tends not to respond to emotionally extreme or overtly positive reviews; instead, it is more likely to engage with detailed expressions of dissatisfaction. Reviews expressing disgust or positive sentiment are more likely to receive a response (Figure 3), with disgust showing a significant positive



Figure 2. Descriptive overview: sentiment tone and App Experience classification share among Spotify reviews.

effect (coefficient = 0.464,  $p < 0.001$ ). In contrast, reviews characterized by anger, joy, or trust tend to be overlooked—joy (coefficient = -0.240,  $p < 0.05$ ) and trust (coefficient = -0.226,  $p < 0.05$ ) both significantly reduce the likelihood of a reply.

### 4.3 Topics: Spotify Prioritizes Problem-Solving

Spotify appears more responsive to critical reviews tied to app performance than to general praise. Approximately 51% of reviews classified as App Experience were negative, compared to just 23% in the Other category. Furthermore, over 75% of all replies were directed to App Experience reviews.

Different topics share broadly similar distributions of emotion scores (Figure 4), with one key exception: the Positive Experience topic tends to reflect high levels of joy and positive sentiment, yet is one of the least likely to receive a reply.

Spotify prioritizes resolving technical access and service issues over addressing feature feedback or acknowledging general satisfaction. Compared to the reference category (Account & Subscription), reviews classified as Music and Playlist Features (coefficient = -0.717,  $p < 0.05$ ), Other (coefficient = -1.031,  $p < 0.001$ ), and Positive Experience (coefficient = -0.915,  $p < 0.01$ ) are significantly less likely to receive a response.

### 4.4 Competitor Mentions: Mentioning Others Doesn't Get Noticed

The ten most frequently mentioned competitors in Spotify reviews are Pandora, YouTube Music, Apple Music, Amazon Music, SoundCloud, Tidal, Deezer, Gaana, Wynk, and Napster. Among the top 5 mentioned competitors, Apple Music, YouTube Music, and Amazon Music tend to show the most joy, positive sentiment, and sadness scores (Figure 5). SoundCloud, YouTube Music, and

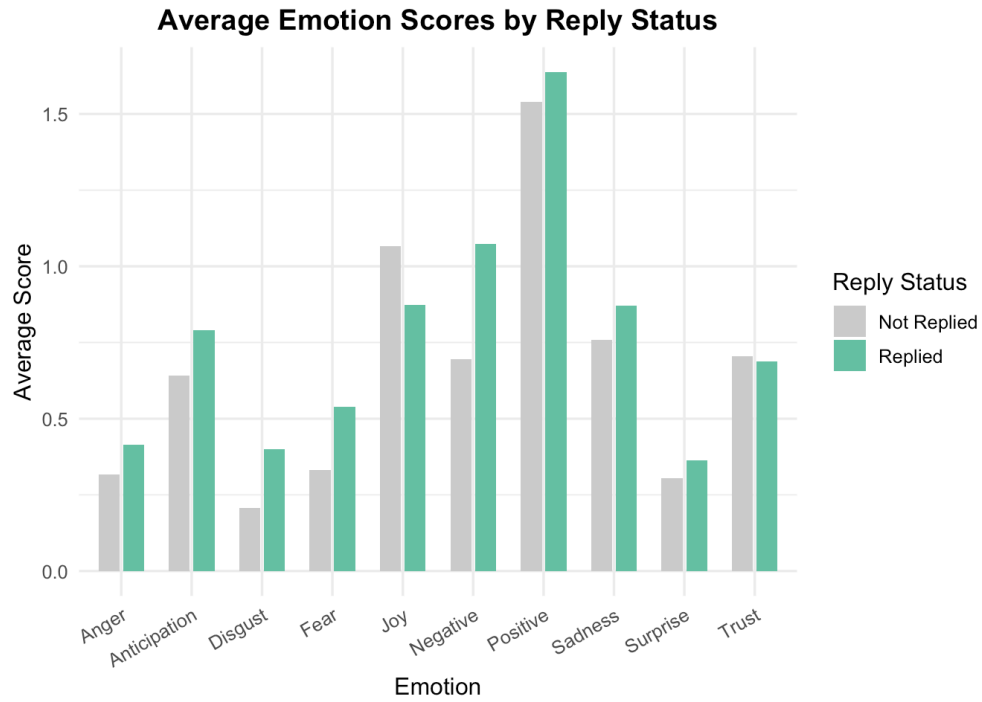


Figure 3. Average emotion scores by reply status: replied vs. unreplied reviews across ten NRC emotion categories.

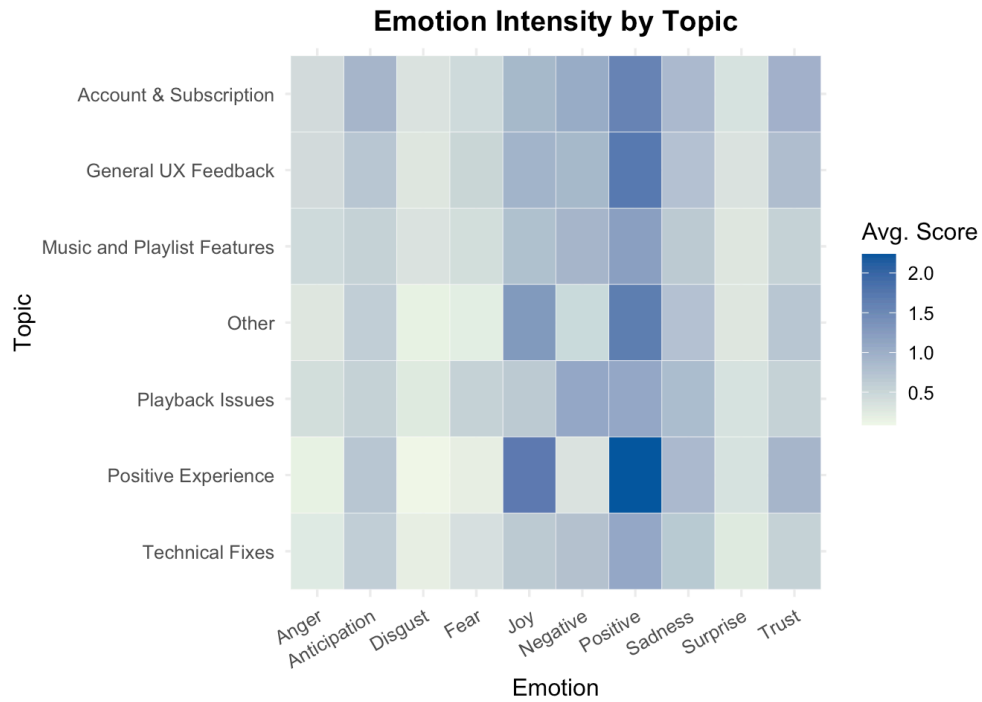


Figure 4. Emotion scores by topic: each LDA-derived topic displays its corresponding average emotion levels using the NRC lexicon.



Amazon Music are often mentioned in negative contexts, whereas Apple Music and Pandora are more frequently cited in positive or neutral reviews).

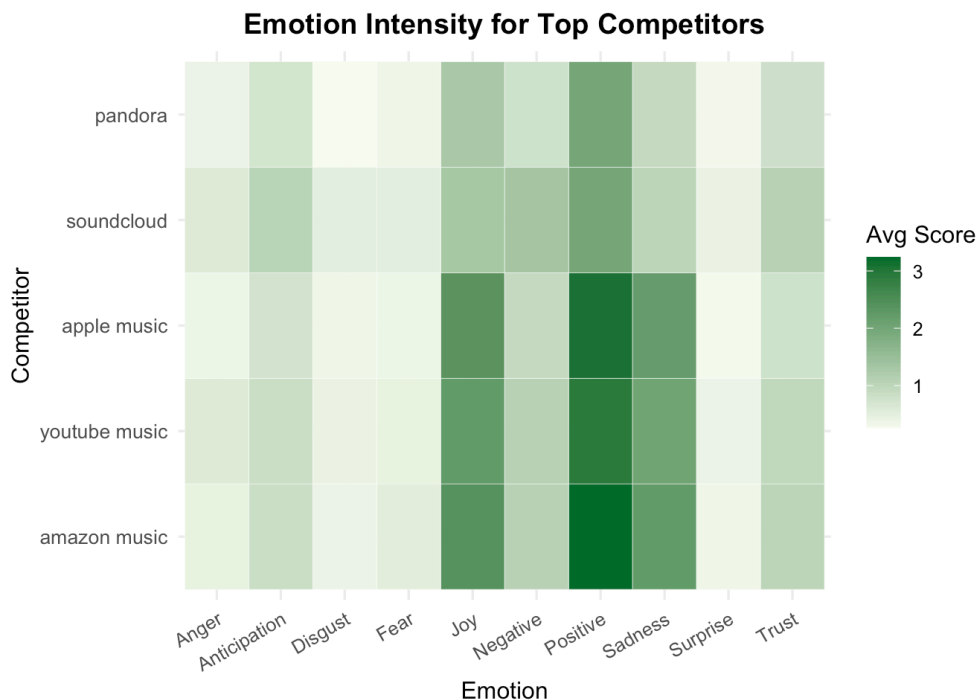


Figure 5. Emotion scores by top 5 competitors

However, Spotify did not explicitly engage with replies containing competitors. According to the logistic regression model, the presence of a competitor mention is not a statistically significant predictor of receiving a reply (coefficient = -0.103,  $p > 0.05$ ).

#### 4.5 Word Count: Spotify Values Detailed User Feedback

Most reviews are relatively concise: the average review contains 32 words, and the interquartile range spans from 14 to 43 words, though some extend up to 738. Reviews that are longer in length are significantly more likely to receive a response: each additional word increases the probability of receiving a reply by approximately 1% (coefficient = 0.010,  $p < 0.01$ ).

#### 4.6 SMOTE Adjustment for Class Imbalance

Spotify’s engagement is systematically focused on complaints, even under balanced sampling. To validate the robustness of these findings, I re-estimated the logistic regression using a SMOTE-balanced dataset to address the low reply rate. The results were largely consistent: emotionally negative tones like disgust and technical topics like Account & Subscription remained strong predictors of reply likelihood (Figure 6; Table 4).

## Comparison of Coefficient Estimates

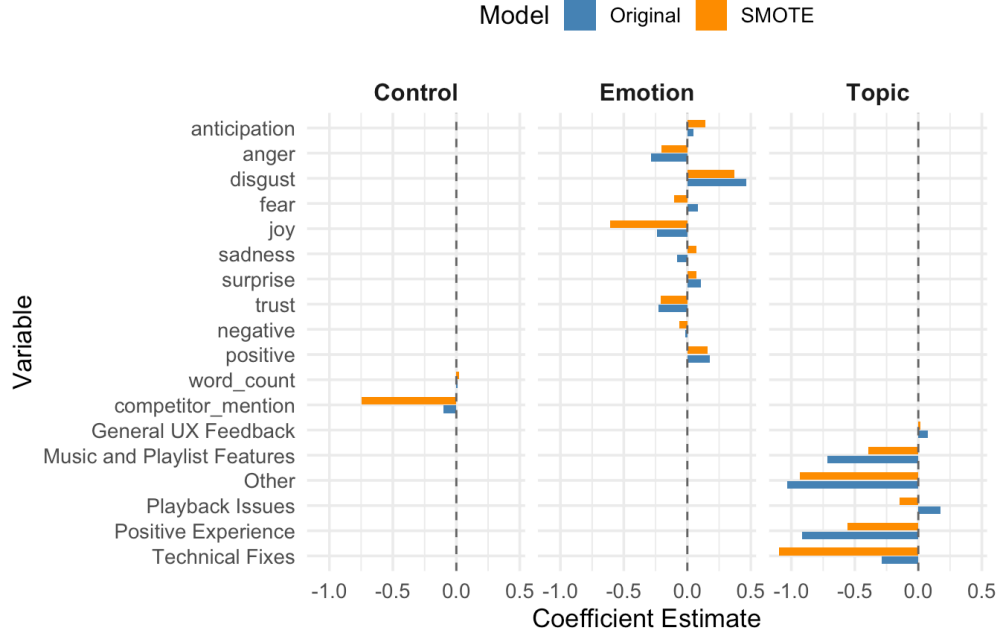


Figure 6. Comparison of Coefficient Estimates

**Table 4. Logistic Regression: Predicting Spotify Replies (Original vs. SMOTE Adjusted)**

| Variable                    | Category       | Original Model    | SMOTE Adjusted Model |
|-----------------------------|----------------|-------------------|----------------------|
| (Intercept)                 | —              | −5.528*** (0.207) | 0.011 (0.021)        |
| anticipation                | NRC Lexicon    | 0.049 (0.090)     | 0.142*** (0.011)     |
| anger                       | NRC Lexicon    | −0.283* (0.137)   | −0.202*** (0.016)    |
| disgust                     | NRC Lexicon    | 0.464*** (0.138)  | 0.371*** (0.017)     |
| fear                        | NRC Lexicon    | 0.086 (0.119)     | −0.106*** (0.014)    |
| joy                         | NRC Lexicon    | −0.240* (0.121)   | −0.609*** (0.014)    |
| sadness                     | NRC Lexicon    | −0.079 (0.107)    | 0.069*** (0.012)     |
| surprise                    | NRC Lexicon    | 0.107 (0.130)     | 0.070*** (0.014)     |
| trust                       | NRC Lexicon    | −0.226* (0.105)   | −0.212*** (0.011)    |
| negative                    | NRC Lexicon    | −0.017 (0.096)    | −0.062*** (0.011)    |
| positive                    | NRC Lexicon    | 0.179* (0.074)    | 0.160*** (0.009)     |
| word_count                  | Other Controls | 0.010** (0.003)   | 0.023*** (0.000)     |
| competitor_mention          | Other Controls | −0.103 (0.426)    | −0.746*** (0.046)    |
| General UX Feedback         | Topic Label    | 0.076 (0.252)     | 0.019 (0.025)        |
| Music and Playlist Features | Topic Label    | −0.717* (0.288)   | −0.395*** (0.025)    |
| Other                       | Topic Label    | −1.031*** (0.250) | −0.934*** (0.022)    |
| Playback Issues             | Topic Label    | 0.176 (0.229)     | −0.148*** (0.024)    |
| Positive Experience         | Topic Label    | −0.915** (0.349)  | −0.560*** (0.027)    |
| Technical Fixes             | Topic Label    | −0.288 (0.255)    | −1.098*** (0.027)    |
| Observations                |                | 60,915            | 121,330              |
| Log Likelihood              |                | −1365.859         | −74247.462           |
| RMSE                        |                | 0.06              | 0.46                 |
| McFadden $R^2$              |                | 0.044             | 0.117                |

Notes: Standard errors are reported in parentheses. \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ .

NRC Lexicon variables are derived from the NRC Emotion Lexicon.

Topic Label reference category is **Account & Subscription**; updated for both models.

Coefficients represent changes in log-odds of receiving a Spotify reply.

## 5 Discussion

Spotify replies to only a small fraction of user reviews and does so selectively, prioritizing longer, more detailed reviews that express dissatisfaction, especially those involving technical or service-related issues. Reviews with emotionally nuanced tones like disgust are more likely to receive a response, while those expressing joy, trust, or general praise tend to be overlooked. Topic modeling and regression results further confirm that Spotify is most responsive to reviews under categories like Account & Subscription. Mentioning competitors such as Apple Music or YouTube Music does not significantly influence reply behavior.

From a business strategy perspective, the analysis identifies key experience themes and sentiment patterns that can inform improvements in Spotify’s app functionality, user satisfaction, and retention efforts. By highlighting which types of reviews prompt engagement—namely those that are detailed, emotionally nuanced, and related to technical service—Spotify can better prioritize customer service efforts and refine its product experience.

From a governance perspective, this study provides transparency into how Spotify manages public engagement through app store replies. It reveals selective responsiveness to certain emotional tones and topics, raising broader questions about fairness, accountability, and communication standards in platform behavior. The use of text-as-data methods contributes to ongoing policy discussions by empirically examining how platform design choices affect user interaction and perceived responsiveness.

Future research could expand this analysis by incorporating data from other platforms (e.g., Apple or YouTube Music) to allow for comparative studies. Further improvements could also include training deep learning models on a larger, more diverse dataset or integrating user metadata to explore demographic patterns in reply behavior. Besides, while reply patterns suggest engagement priorities, further work is needed to fully assess transparency and policy implications.

### 5.1 Limitations

While the findings offer clear patterns, several limitations remain. First, TF-IDF outperformed word embeddings in classification, which may reflect insufficient embedding tuning or the brevity of most reviews.

Second, topic modeling and classification outputs showed some conceptual overlap, limiting interpretability. Enhanced preprocessing or alternative models could improve topic separation.

Third, although the number of retained duplicate reviews is small (270), their short and repetitive nature may reduce analytical quality. Word clouds also revealed minor stemming issues, such as splitting “song” and “songs.”

Finally, fewer than 3.5% of reviews received a reply, significantly limiting the statistical power of the regression model.

### 5.2 R Packages Used

- **Data Cleaning and Transformation:** `tidyverse`, `janitor`, `textclean`, `hunspell`

- **Text Preprocessing:** `tidytext`, `quanteda`, `textmineR`, `quanteda.textstats`
- **Topic Modeling:** `topicmodels`, `broom.mixed`
- **Classification:** `caret`, `glmnet`, `e1071`, `naivebayes`
- **Visualization:** `ggplot2`, `patchwork`, `wordcloud`, `RColorBrewer`
- **Model Reporting:** `pscl`, `modelsummary`, `gridExtra`
- **SMOTE Handling:** `smotefamily`

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